COMMON COURSE ASSIGNMENT FORMAT

A. Assignment Information

Particular	Details
Department:	AI&DS
Programme:	B.Tech
Course Code & Course Name:	DSA0606 - Data Handling and Visualization
Academic Year / Batch:	2026-2027/2023
Faculty Name:	Dr. Karthikeyan. L
Assignment Title:	Social Media Sentiment & Engagement Data Visualization
Date of Issue:	30.08.2026
Date of Submission:	31.08.2026
Maximum Marks:	100
Course Outcome(s) – CO:	CO1-CO5 (Visualising data, types of visualisations, distributions & proportions, relationships & time series, uncertainty & effective design principles)
Bloom's Taxonomy Level:	L3 Apply, L4 Analyse, L5 Evaluate
SDG Mapping (SDG 1-17):	SDG 9 (Industry, Innovation & Infrastructure); SDG 8 (Decent Work & Economic Growth)
Industry / Societal Relevance:	Enables marketing teams to convert unstructured social-media streams into timely, evidence-based campaign decisions

B. Assignment Problem / Challenge

Problem Statement

- Problem / Case / Design Challenge:

A brand's social accounts have accumulated 420 posts across five platforms, each with text, hashtags, topic, user type metadata and engagement metrics. The team must handle the unstructured (free text) and semi structured (hashtag arrays, timestamp strings) components, clean and enrich the dataset, and create visualizations that help content strategists understand how sentiment and engagement vary by platform, topic, time and user type and to anticipate viral complaints, campaign lifts and optimal posting windows.

Requirements and Constraints

- Inspect and understand the structure and quality of the social media dataset (20 fields, 5 platforms, 8 topics).
- Identify and handle missing, duplicate, inconsistent, and invalid records (timestamps, engagement outliers, text noise).
- Clean unstructured text (URLs, @mentions, punctuation, case) and normalize semi structured fields (hashtag arrays, timestamps).
- Score sentiment via a transparent lexicon pipeline (tokens → lexicon match → score in [-1,+1] label).
- Create clear, filter-driven visualizations for strategic decision making (8 charts + table + playground).
- Since no personally identifying information is included, the analysis must focus on aggregate patterns and operational insights.

Student Work

Problem Understanding and Formulation

- What is the problem?
  Organizations increasingly rely on social media not only as a broadcast channel but as a real time sensor of public attitude. Temporal and behavioral context strongly influences social performance, requiring strategists to recognize recurring platforms, topic and temporal patterns without premature modeling.
- What are the expected outcomes?
  To develop practical recommendations for content scheduling, platform investment, and early warning triage using visualizations.
- What information/data is available?
  A synthetic but realistic dataset containing 420 anonymized post records spanning 2026-03-01 to 2026-08-31. Table 1 outlines the core variables:

Variable	Description	Role in Analysis

Post ID	Synthetic identifier (P0001 P0420), chronological.	Record tracking & ordering; duplicate checking
Platform	Publishing platform: Twitter, Instagram, Facebook, YouTube, LinkedIn.	Platform comparison; baseline engagement modeling
Timestamp / Hour	Publication time (YYYY-MM-DD HH:MM: SS) and derived hour 0-23.	Temporal & hourly trend analysis
Text	Raw post content with hashtags, @mentions and URLs.	Tokenization & sentiment scoring
Sentiment Score/Label	Lexicon derived score [-1,+1] and label (positive/neutral/negative).	Sentiment distribution & correlation
Engagement Metrics	Likes, shares, comments, views, engagement (sum) and engagement rate.	Performance ranking & scatter analysis
Hashtags/Topic / User Type	Hashtag array, topic (8 categories), user type (Regular/Influencer/Brand/Verified).	Topic sentiment & user type lift analysis

Table 1: Core variables in the social media dataset.

- What assumptions/constraints must be satisfied? The analysis is designed as a decision support exercise rather than a production sentiment classifier.

Application of Course Knowledge

Exploratory Data Analysis (EDA) provides a systematic approach combining data cleaning, descriptive statistics, grouping, comparison, and visualization. Table 0 maps the course outcomes to the concepts applied:

S.No.	Concept	Description
CO1	Text Cleaning	Cleansing unstructured post text by removing URLs, mentions, noise and normalizing case.
CO2	Sentiment Classification	Classifying posts into positive, neutral and negative using lexicon based scoring.
CO3	Engagement Analysis	Comparing likes, shares and comments across platforms, topics and user types.
CO4	Temporal Visualization	Visualizing engagement and sentiment patterns

		across hours, days and campaign spikes.
CO5	Interactive Dashboard	Enabling filtered exploration of sentiment & engagement patterns in a static app.

Table 2: Course Outcome mapping for DSA 0606: from data handling to visualization.

Solution / Design / Methodology

Data Cleaning and Preprocessing: Missing timestamps that could not be recovered were excluded, and missing sentiment scores were recomputed via the lexicon pipeline rather than imputed. Text cleaning lower-cased content, stripped URLs and @mentions, isolated "#" tokens, replaced punctuation with spaces, and collapsed whitespace.

Categorization: Each post's lexicon score was mapped to a label using thresholds: score > +0.25→ positive, score < -0.25 → negative, otherwise neutral. Engagement was computed as likes+shares+comments.

Visualization Plan: Table 2 maps each chart to a distinct analytical question:

Visualization	Purpose	Strategic Use
Donut chart	Show sentiment distribution	Set per topic sentiment baselines & alerts
Timeline area/line	Reveal engagement spikes over time	Time post campaign reviews & retrospectives
Stacked bar (topic sentiment)	Show sentiment composition per topic	Tailor topic specific messaging & moderation
Grouped bar (platform)	Compare platform engagement components	Allocate creative/budget by platform
Scatter (sentiment vs engagement)	Expose correlation & viral outliers	Triage high engagement complaints proactively
Horizontal bar (hashtags)	Rank tag salience	Optimize tag lexicon for campaigns vs support
Dual axis line+bar (hourly)	Compare hourly efficiency vs volume	Schedule posts for peak efficiency windows
Bar (user type)	Compare influencer lift	Decide partnership & amplification strategy

Table 2: Visualization plan: each chart maps to a distinct analytical question.

Pseudocode:

ALGORITHM SocialMediaSentimentPipeline

1. LOAD synthetic posts (n=420, seed=42)

2. FOR each post DO
3. CLEAN text (lowercase, strip URLS, @mentions, punctuation)
4. NORMALIZE hashtag array and platform/topic labels
5. COMPUTE total engagement = likes + shares + comments
6. COMPUTE sentiment_score = Lexicon Polarity (text)
7. ASSIGN label using thresholds (+0.25/-0.25)
8. END FOR
9. DERIVE hour, date, day_of_week, campaign_peak flags
10. GENERATE grouped summaries and eight chart views
11. EMBED data + Chart.js views into single static HTML file

Use of Modern Tools

- Python + pandas: For data generation, cleaning, sentiment scoring, aggregation.
- Matplotlib: To generate print grade figures for the report.
- HTML/CSS/vanilla JavaScript + Chart.js (CDN): For rendering a single file offline interactive dashboard (SocialPulse app).
- Git/GitHub: For version control and final repository hosting.

Results and Validation

- Sentiment distribution: Positive 40.2% (169), Neutral 29.8% (125), Negative 30.0% (126), with a Mean score ≈ +0.06.
- Platform engagement: Instagram and YouTube lead on average engagement; LinkedIn trails.
- Topic sentiment: Marketing Campaign and Product Launch skew positive; Customer Service is negative dominant (~50% negative).
- Temporal peaks: Four injected campaign peaks (Apr 15, May 20, Jun 10, Jul 18) appear as clear engagement leaders, with evening hours (18:00 19:00) delivering the highest per post engagement.
- User type lift: Influencer posts outperform Regular posts by approximately 80% on average engagement.

Visualizations Generated:

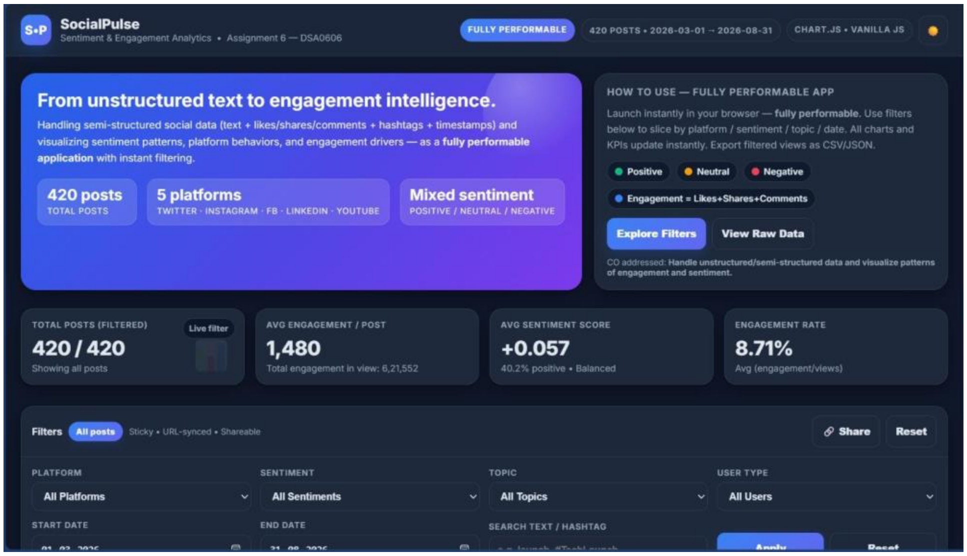

Figure: Social Media Sentiment Engagement Dashboard

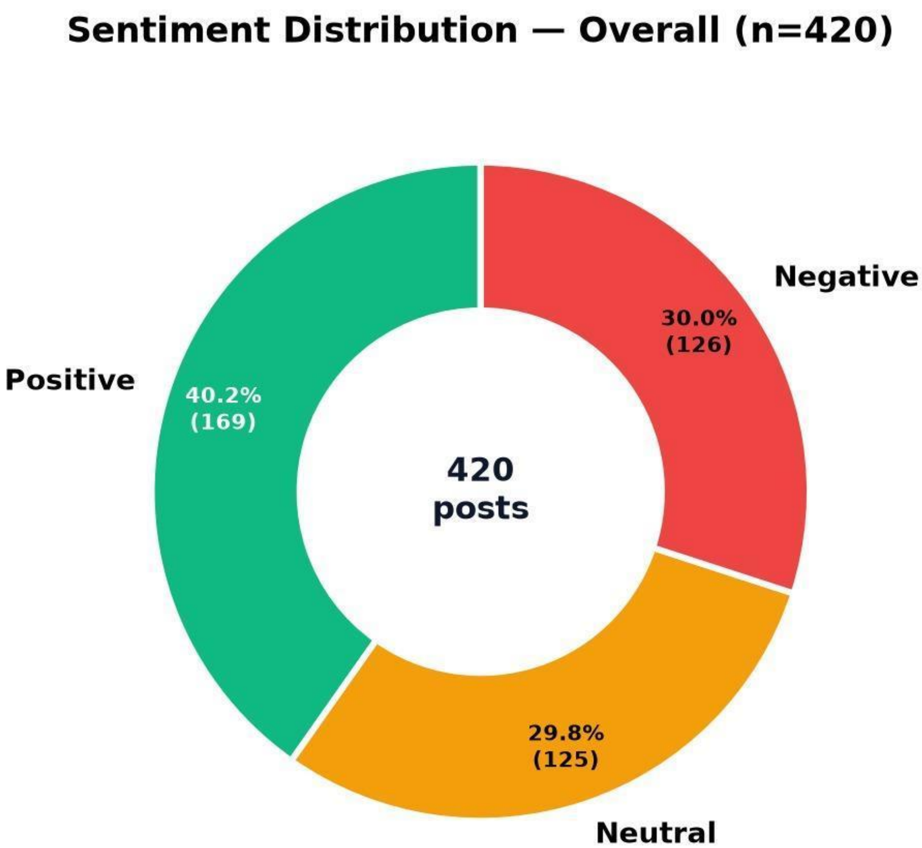

Figure 1. Overall sentiment distribution (n=420)

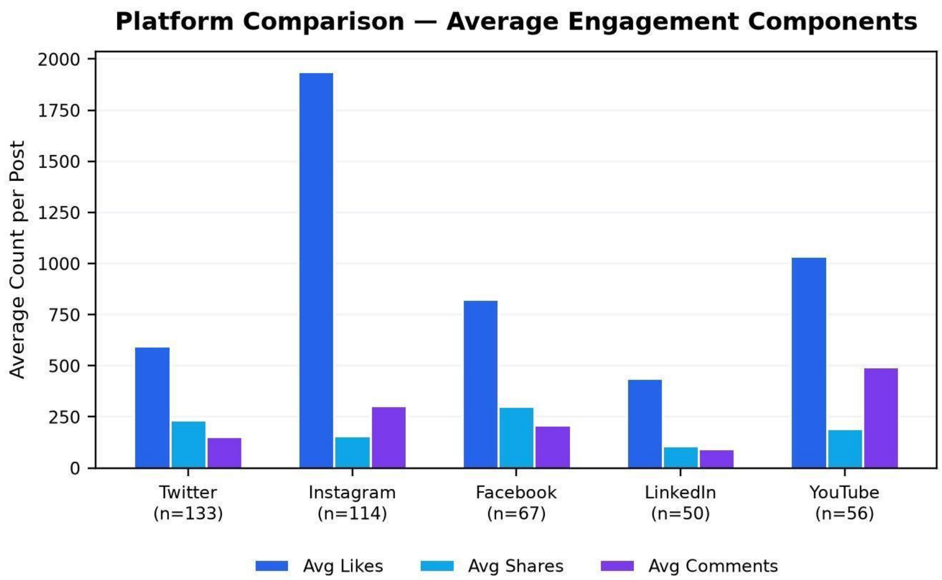

Figure 2. Average likes, shares and comments per post by platform

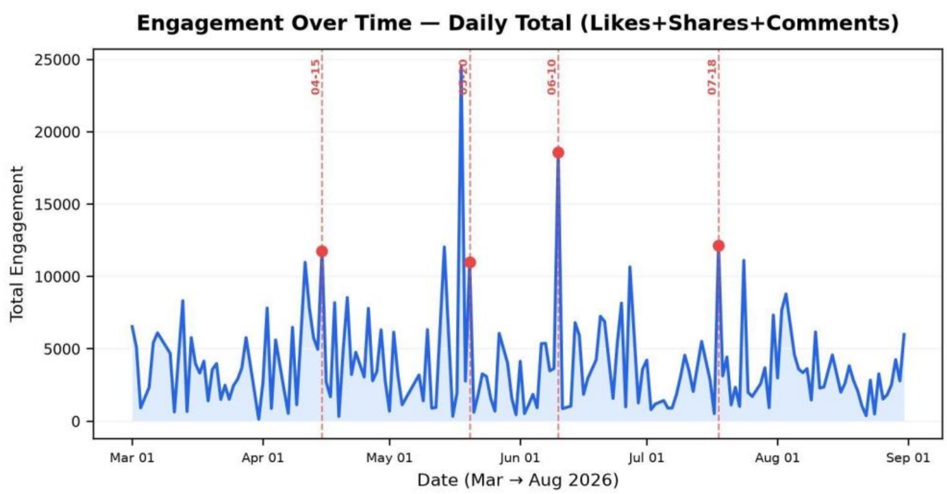

Figure 3. Total engagement over time; campaign peaks marked

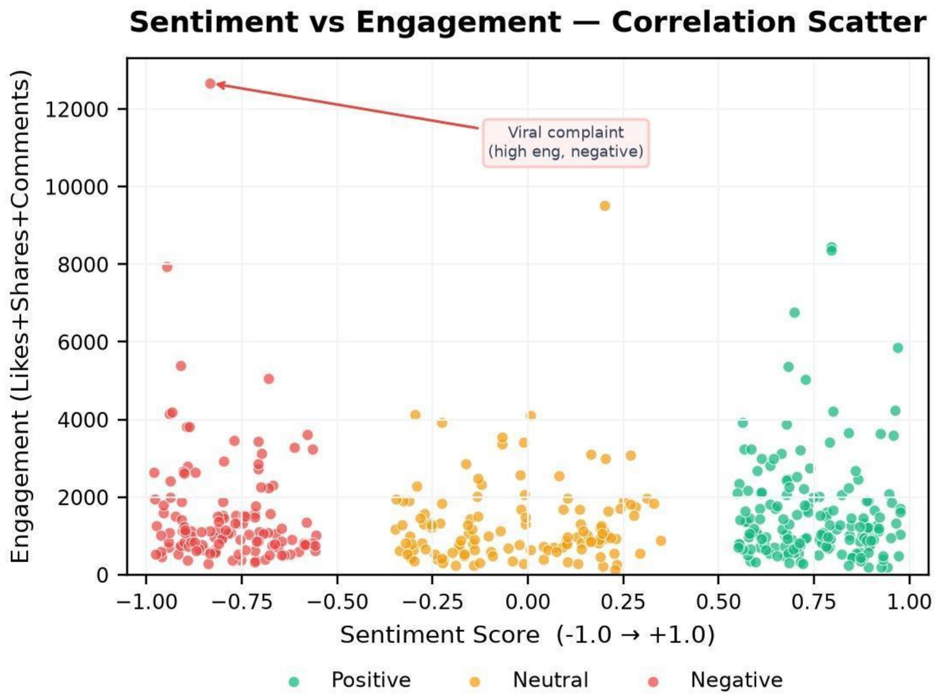

Figure 4. Sentiment score vs engagement; viral tails visible

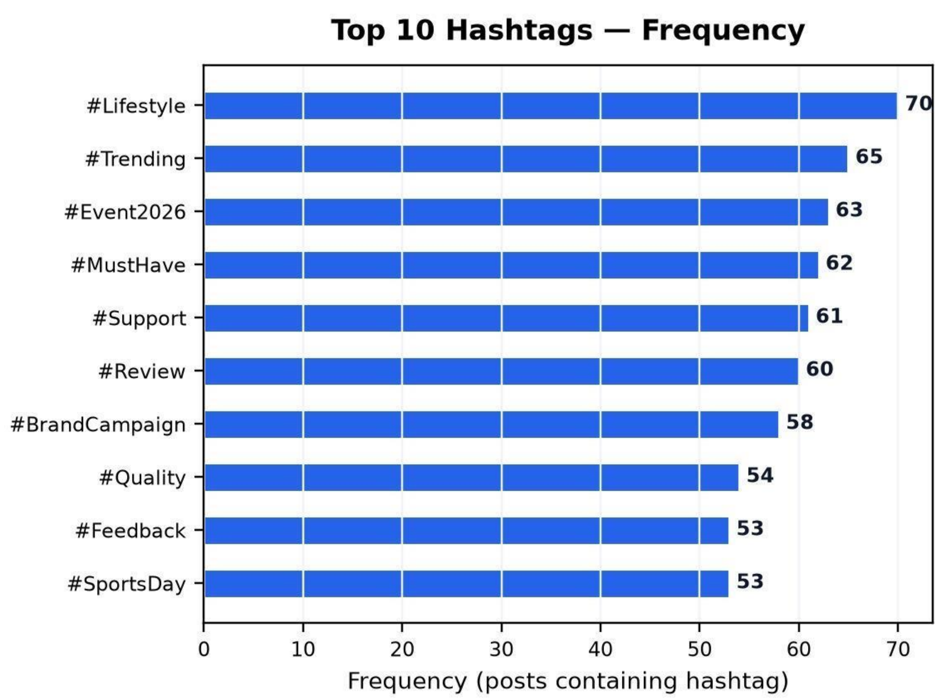

Figure 5. Frequency of hashtag array elements

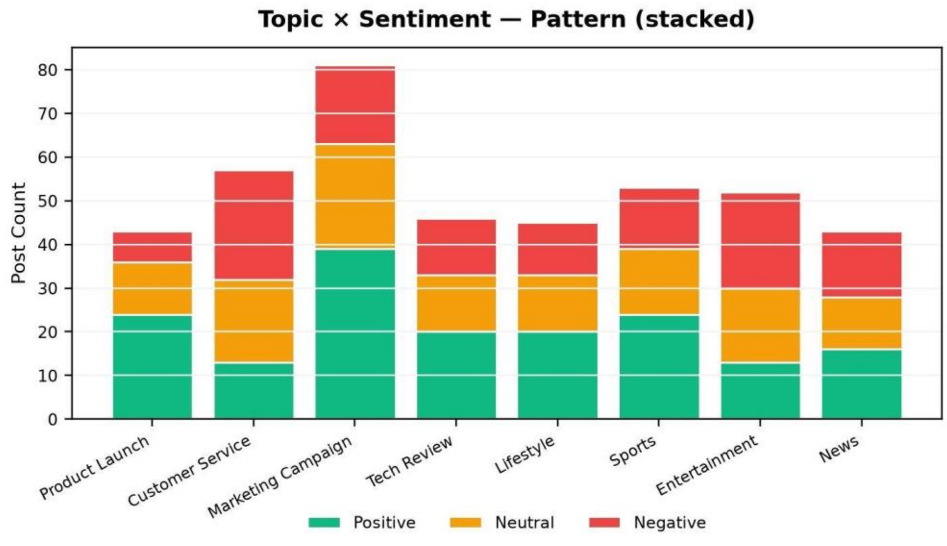

Figure 6. Sentiment composition per topic

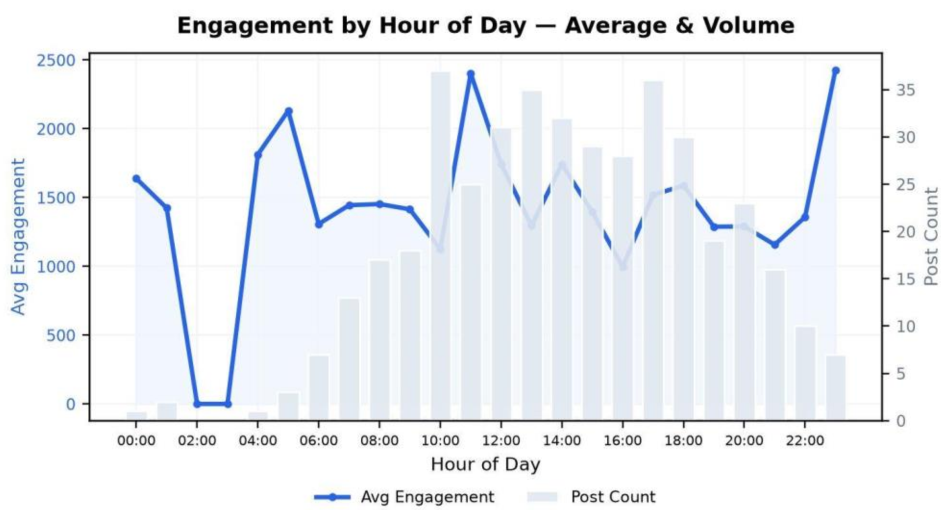

Figure 7. Avg engagement (line) and post count (bars) by hour

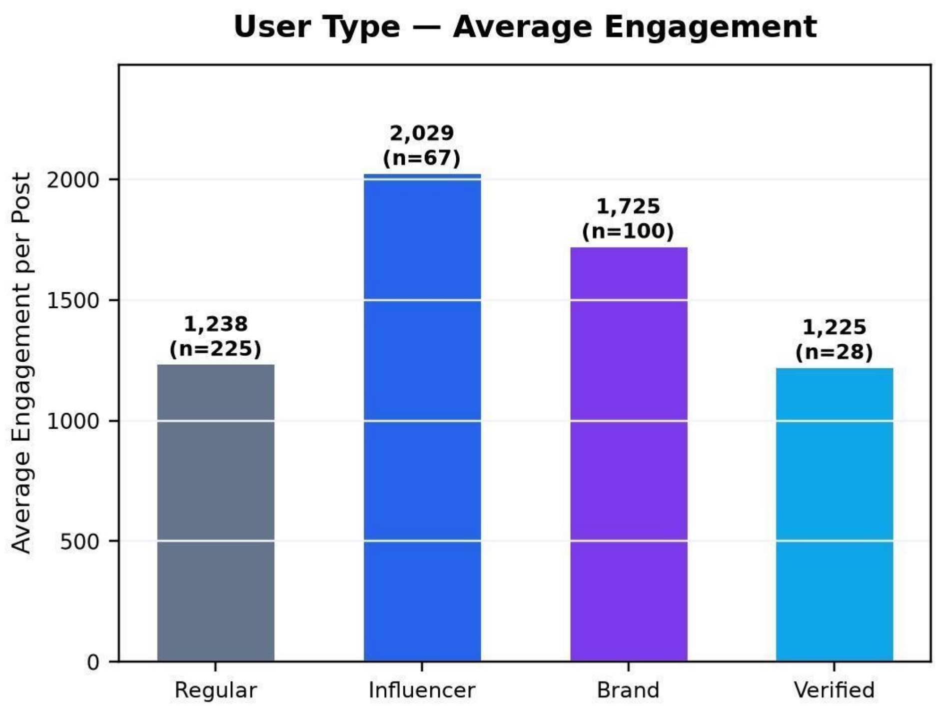

Figure 8. Average engagement per post by user type

Analysis and Engineering Decision

- Strategy Trade-offs & Decisions: If historical data indicates that engagement spikes during campaign windows and that those spikes are strongest on Instagram and YouTube, future campaign launches can be aligned with similar calendar positions and visual assets prioritized for those platforms.
- Scheduling Decisions: Evening hours (18:00 19:00) consistently deliver higher per post engagement and should be preferred for high value posts.

Broader Considerations

- Professional Responsibility & Moderation: High engagement negative posts (the viral tail) should trigger proactive moderation triage rather than algorithmic suppression.
- Limitations: A rule based lexicon cannot capture sarcasm, negation or domain jargon. Single language (English) masks code switching common in multilingual markets, and historical patterns cannot guarantee future behavior.

Conclusion

This project demonstrates how exploratory data analysis can be applied to synthetic but realistic social media data to support practical strategic decision making. By cleaning unstructured text, normalizing semi structured fields, scoring sentiment and engineering engagement and temporal features, the team transformed raw posts into understandable, actionable summaries. The analysis identifies which platforms deliver the highest engagement, which topics skew positive or negative, when engagement peaks by hour and campaign phase, and which user types and posts constitute the viral tail.

Student Reflection

- What did you learn from this assignment beyond what was directly taught in the classroom?

Through this assignment, we gained hands-on experience bridging the gap between unstructured social media text and actionable business intelligence. We learned how to effectively handle noisy data—such as mixed-case text, emojis, and erratic timestamps—and transform it into a structured format suitable for analysis. Building the lexicon-based sentiment pipeline taught us the practical nuances of text normalization and polarity scoring. Furthermore, developing the interactive dashboard provided valuable insights into translating backend data into accessible, filter-driven visualizations for non-technical stakeholders.

- If you were given additional time/resources, what would you improve and why?

If given additional time and resources, we would replace the rule-based lexicon with a machine learning or transformer-based NLP model (such as BERT or RoBERTa) to better capture sarcasm, context, and multilingual code-switching. Additionally, we would connect the static dashboard to a live social media API to analyze real-time streaming data rather than relying on a static historical dataset.

Individual Contribution

S.No.	Student Name	Individual Contribution
1	BALLANI VENKATA MANOJ	Project Lead, Dashboard Architecture & HTML/JS (Chart.js) Integration.
2	Priyanka G	Lexicon Sentiment Pipeline Development & Data Categorization.
3	DHAYALAN B	Temporal Feature Engineering & Engagement Metrics Calculation.
4	Kishore M P	Data Cleaning, Quality Assessment & Technical Documentation.
5	SANIA HERBERT	Led the data preprocessing pipeline using Python and Pandas, and generated all print-grade statistical visualizations using Matplotlib.

References

- Adekoya-Cole, T., Fernando, S., Maroju, A., Samal, C. K., Aggarwal, S., & Brahma, B. (2026). Building a dynamic opinion dashboard to categorize tweets for real-time sentiment analysis. International Journal of Information Technology, 18(1), 5-11.

- Al-Montaser, M. A., et al. (2024). Sentiment analysis of social media data: Business insights and consumer behavior trends in the USA. ResearchGate preprint.
- Kurtz, G. B., Carraro, S. de P., Teixeira, C. R. G., Bandeira, L. D., Müller, B. L., Tietzmann, R., Silveira, M., & Manssour, I. (2025). Stream Vis: An analysis platform for YouTube live chat audience interaction, trends and controversial topics. Proceedings of the 27th International Conference on Enterprise Information Systems (Vol. 2, pp. 630-640). SciTePress.
- Masson, et al. (2024). TextBI: An interactive dashboard for visualizing multidimensional NLP annotations in social media data. Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics: System Demonstrations (pp. 1-10).
- PLOS ONE. (2026). Predicting social media post popularity using multimodal deep learning: Insights from content features. PLOS ONE, 21(8), e0356612.
- IEEE Transactions on Visualization and Computer Graphics. (2025). Causality-based visual analytics of sentiment contagion in social media topics. IEEE Transactions on Visualization and Computer Graphics.
- Zhao, et al. (2024). A comprehensive review of visual-textual sentiment analysis from social media networks. Journal of Computational Social Science, 7, 1235-1278.

Common Assessment Rubric

Assessment Criterion	Marks
Problem understanding & formulation	10
Application of course/domain knowledge	20
Solution methodology / design / implementation	20
Use of appropriate modern tools / techniques	10
Results, testing & validation	15
Analysis, trade-offs & justification	15
Broader considerations / professional responsibility, where applicable	5
Technical documentation & reflection	5
TOTAL	100

CO–PO–Assessment Mapping

Assessment Component	CO	PO(s)	Bloom's Level	Marks
Problem formulation			L3/L4	10
Application of knowledge			L3/L4	20
Solution / Design / Implementation			L4/L5/L6	20
Modern tool usage			L3/L4	10
Validation			L4/L5	15
Analysis & justification			L4/L5	15
Broader considerations			L4/L5	5
Documentation & reflection			L3/L4	5

Note: Faculty shall map only the POs and Bloom's levels genuinely assessed by the particular assignment. Every assignment is not required to map to every PO.

H.Faculty Evaluation & Continuous Improvement CO Attainment

Performance Level	Criterion
Level 3	$\geq$ 70%
Level 2	60–69%
Level 1	50–59%
Level 0	< 50%

Target CO Attainment:

Actual CO Attainment:

Faculty Analysis
Areas in which students performed well:

Areas requiring improvement:

Corrective / Improvement Action:

Action to be implemented in the next offering:

Documents to be Maintained
The department/course faculty shall maintain:

- Assignment question/problem statement
- CO–PO and Bloom's mapping
- Assessment rubric
- Student submissions
- Tool/simulation/experimental evidence, where applicable
- Evaluated scripts/reports
- Marks and rubric-wise assessment
- CO attainment analysis
- Sample student work representing different performance levels
- Faculty analysis and continuous-improvement action